Shared-Target Rigidity and Dominance Energy in Fixed-Shift Determinant Fibers:

Sharp Phase Envelopes, Compressed Census Identities, and a Loss-One Coherence Barrier

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July 2026

Abstract

For the matched fixed-shift coefficient of TPC-32, natural determinant-energy survival is equivalent to survival of the literal post-bin ℓ^1 mass, but neither lower bound is known. We make two strict reductions inside this open branch.

First, for each determinant fiber let R_n be its raw absolute mass, $r_{\max,n}$ its largest native modulus, and $d_n = (2r_{\max,n} - R_n)_+$. We prove that

$$D_{\mathrm{dom}} := \sum_n d_n^2$$

is the exact minimum determinant energy among all packets with the same fiber partition and native moduli but freely variable phases. The corresponding total equal-determinant coherence has the complete sharp interval

$$\left[D_{\mathrm{dom}} - E_{\mathrm{trip}},\ \sum_n R_n^2 - E_{\mathrm{trip}} \right].$$

This improves the earlier route through a global post-bin mass certificate and gives an exponent-exact dominant-fiber sufficient condition.

Second, we use the primitive row geometry to prove a rigidity theorem. On canonical parent records, one normalized determinant together with the two ordered affine targets identifies the native row–orbit key uniquely. More generally, the graph joining equal-determinant keys that share any complete target has subpower maximum degree. Its ordered coherence admits an exact compressed formula in terms of four target-group sums, without materializing pairs, and satisfies

$$|C_{\mathrm{tar}}| \le X^{o(1)} E_{\mathrm{trip}} \le X^{-1+o(1)} Q^3 J^2.$$

Consequently, conditional on any future determinant-energy lower bound with fixed loss $\lambda_D < 1$, shared-target coherence is $|C_{\mathrm{tar}}| = o(D)$ and cannot carry the main term. The result removes one candidate pairing route but leaves shared-divisor pairings with distinct targets, other determinant-preserving structures, and the unpaired signed remainder open. All new asymptotic statements are interfaces or upper bounds; no post-bin mass or energy lower bound and no new growing fixed-shift arithmetic estimate is proved.

Keywords: fixed shift; determinant fiber; native key; target collision; dominance energy; coherence graph; exact census.

Literal-frame convention. Every physical statement retains one prescribed nonzero shift h_0, canonical parent native keys, exact active targets, the actual post-bin coefficient, and the original global normalization. No statement is specialized to $h_0 = 2$.

1 Introduction

Let $A_X(n)$ be the literal signed normalized-determinant coefficient obtained after the three TPC-32 matched cutoff channels and all native contributions in each determinant fiber have been aggregated [4]. TPC-83 proves, using the actual support and singleton envelopes, that natural post-bin mass and natural determinant energy are equivalent:

$$\|A_X\|_1 \geq X^{-o(1)} N_{0,X} \quad \Longleftrightarrow \quad \|A_X\|_2^2 \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}.$$

This identifies the correct variable but supplies neither lower bound [6].

TPC-84 then opens the coefficient into canonical native determinant fibers. It proves the sharp resultant interval from the native moduli and the exact identity

$$D = E_{\text{trip}} + C_{\text{eq}}.$$

Since E_{trip} lies at most on the loss-one scale, a future energy lower bound with fixed loss below one must be carried by positive off-diagonal equal-determinant coherence [7]. The paper lists exact duplicates, determinant-preserving symmetries, shared targets or divisors, and an unpaired remainder as candidate classes, while deliberately assigning no unproved sign or size to them.

The present paper performs the next two exact steps.

1. We square the sharp fiber resultant before summing. This gives the optimal modulus-only energy certificate
$$D_{\text{dom}} = \sum_n (2r_{\max,n} - R_n)_+^2.$$
It is never weaker than first summing the defects and then applying Cauchy–Schwarz with the entire support, and it can be strictly stronger when the surviving defect is concentrated.

2. We show that a same-determinant shared-target graph is too sparse to supply the required coherence. The key is not a heuristic independence claim. It is an exact parametrization: fixing a determinant and one affine target leaves at most a divisor-bounded number of canonical parent keys. This yields a subpower degree and hence a loss-one absolute bound for the whole shared-target class.

The second result is stronger than saying that no positive shared-target main term has been found. It proves that even perfect phase alignment inside that declared class cannot reach any determinant-energy scale with fixed loss below one. Thus this particular door is closed. The conclusion is class-local: sharing a selected divisor without sharing a complete target is a different relation and remains open.

1.1 Main results

The finite geometry gives the exact free-phase range

$$D \in \left[\sum_n d_n^2, \; \sum_n R_n^2 \right].$$

Subtracting the fixed triplet diagonal gives the exact free-phase range of C_{eq}. A dominant share of the raw mass therefore supplies a deterministic energy lower bound with an explicit exponent ledger.

For the arithmetic rigidity, write one canonical parent as

$$t = (\alpha, \gamma, j), \qquad x_t = m_\alpha j + h_0, \qquad y_t = m_\gamma j + h_0,$$

and

$$\nu(t) = \frac{m_\alpha - m_\gamma}{(x_t, y_t)}.$$

The physical orbit has $j > 0$, and the row mask has $m_\alpha \neq m_\gamma$. We prove:

(i) the composite key $(\nu(t), x_t, y_t)$ identifies t;

(ii) two distinct equal-determinant keys share at most one target, even if left–right cross-sharing is allowed;

(iii) the target-sharing correlation is an exact sum of compressed left, right, and cross group statistics;

(iv) every target group has divisor-bounded cardinality; and

(v) the resulting coherence is at most the loss-one scale.

The target-group formula also provides an audit invariant for a future native-key census. It requires only sorting by exact integer keys and does not create a quadratic pair table.

1.2 Claim discipline

The sharp phase results are L0 finite algebra. Their placement on the literal post-bin coefficient and the target-collision bound are L1 results because they preserve the fixed nonzero h_0, canonical parent keys, active target support, and global normalization. There is no new L2 theorem: no lower bound for raw mass, post-bin mass, determinant energy, or the distinguished zero mode is proved. In particular, no sieve-parity, prime-pair, or twin-prime conclusion is claimed.

2 The literal native packet

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$. We work on the high-$\beta$ matched packet

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}, \qquad C_X = \lfloor J_X \rfloor, \tag{1}$$

with

$$N_{0,X} = J_X Q_X^2, \qquad Q_X J_X \asymp_{\mathscr{D}} X. \tag{2}$$

The symbol $\mathscr{D}$ contains fixed support, smoothness, residue, orientation, and positive-margin data. Dependence on these data and on h_0 is suppressed.

2.1 Canonical parent rule

A canonical parent native key is

$$t = (\alpha, \gamma, j).$$

The rows are ordered and off-diagonal. Their slopes satisfy

$$m_\alpha \asymp Q_X, \qquad m_\gamma \asymp Q_X, \qquad m_\alpha \neq m_\gamma,$$

and the primitive orbit support has

$$j \asymp_{\mathscr{D}} J_X > 0.$$

In the TPC-32 row construction the slope is represented as $m_\alpha = \ell_\alpha d_\alpha$; the prime-size separation makes the integer slope determine the row key uniquely [2–4]. We use only this uniqueness, not the displayed mnemonic decomposition.

Definition 2.1 (Canonical parent archive)**.** The set $\mathcal{T}_X$ contains exactly one record for each mathematical parent triple (α, γ, j). Matched-shell channels, cutoff levels, divisor openings, and other computational children are first inverse-aggregated into their parent. Their sum is the literal coefficient $u_t \in \mathbb{C}$. A computational copy is not a second native key, while a genuinely distinct parent retains its distinct row or orbit identity.

Remark 2.2 (Why the premise is explicit). Without definition 2.1, copying one archive row K times would create an arbitrary target-group multiplicity and an arbitrary off-diagonal "coherence" without changing the physical formula. Every uniqueness and degree statement below is therefore made only after canonical parent aggregation. This is the same inverse-multiplicity rule required by the TPC-84 census schema.

For $t = (\alpha, \gamma, j)$, define the ordered targets, full content, and canonical determinant

$$\begin{aligned} x_t &= m_\alpha j + h_0, \qquad & y_t &= m_\gamma j + h_0, \\ c_t &= (x_t, y_t), \qquad & \nu(t) &= \frac{m_\alpha - m_\gamma}{c_t}. \end{aligned} \tag{3}$$

Primitive target geometry gives

$$c_t \mid m_\alpha - m_\gamma, \qquad (c_t, m_\alpha m_\gamma h_0) = 1,$$

so $\nu(t) \in \mathbb{Z} \setminus \{0\}$ [2, 3]. Both targets are positive and $\asymp_{\mathscr{D}} X$.

2.2 Fibers and energies

For $n \in \mathbb{Z} \setminus \{0\}$, put

$$F_n = \{t \in \mathcal{T}_X : \nu(t) = n\}, \qquad A_X(n) = \sum_{t \in F_n} u_t. \tag{4}$$

Define

$$\begin{aligned} R_X &= \sum_{t \in \mathcal{T}_X} |u_t|, \quad & M_X &= \sum_n |A_X(n)|, \\ E_{\mathrm{trip},X} &= \sum_{t \in \mathcal{T}_X} |u_t|^2, \quad & D_X &= \sum_n |A_X(n)|^2. \end{aligned} \tag{5}$$

The ordered equal-determinant coherence is

$$C_{\mathrm{eq},X} = \sum_n \sum_{\substack{t,t' \in F_n \\ t \neq t'}} u_t \overline{u_{t'}}. \tag{6}$$

Opening $|A_X(n)|^2$ gives the exact identity

$$\boxed{D_X = E_{\mathrm{trip},X} + C_{\mathrm{eq},X}.} \tag{7}$$

The imported literal envelopes are

$$\# \operatorname{supp} A_X \ll Q_X, \tag{8}$$

$$R_X \ll X^{o(1)} N_{0,X}, \tag{9}$$

$$E_{\mathrm{trip},X} \ll X^{o(1)} N_{0,X}. \tag{10}$$

The last bound and (2) imply

$$E_{\mathrm{trip},X} \ll X^{-1+o(1)} Q_X^3 J_X^2. \tag{11}$$

These are upper bounds only. In particular, (9) is not a raw-mass lower bound.

3 The sharp dominance-energy certificate

We first work with an arbitrary finite complex packet partitioned into fibers. For each fiber define

$$R_n = \sum_{t \in F_n} |u_t|, \qquad r_{\max,n} = \max_{t \in F_n} |u_t|, \qquad d_n = (2r_{\max,n} - R_n)_+, \tag{12}$$

with all three quantities zero for an empty fiber. Put

$$D_{\mathrm{dom}} := \sum_n d_n^2, \qquad D_{\mathrm{align}} := \sum_n R_n^2. \tag{13}$$

TPC-84 proves that, with the fiber moduli fixed and phases free, the resultant magnitude in fiber n fills the complete interval $[d_n, R_n]$ [7]. Squaring before summing gives the energy statement missing from the earlier mass route.

Theorem 3.1 (Sharp free-phase determinant-energy interval)**.** *Fix a finite fiber partition and all native moduli $(|u_t|)_{t\in\mathcal{T}}$. Allow the phases of the native coefficients to vary independently. Then the set of attainable determinant energies is exactly*

$$\boxed{\left\{\sum_n \left|\sum_{t\in F_n} u_t\right|^2\right\} = [D_{\mathrm{dom}}, D_{\mathrm{align}}].} \tag{14}$$

In particular, every actual phase assignment satisfies

$$\boxed{D \geq D_{\mathrm{dom}} = \sum_n d_n^2.} \tag{15}$$

Proof. For each nonempty fiber, the attainable resultant magnitudes form $[d_n, R_n]$; hence the attainable squared magnitudes form $[d_n^2, R_n^2]$. The phases of disjoint fibers may be chosen independently. A Minkowski sum of finitely many real intervals is the interval whose endpoints are the sums of their endpoints. This proves (14) and its lower bound. □

Corollary 3.2 (Sharp free-phase coherence interval)**.** *With the same fixed moduli, E_{trip} is fixed and the complete attainable interval of ordered equal-fiber coherence is*

$$\boxed{C_{\mathrm{eq}} \in [D_{\mathrm{dom}} - E_{\mathrm{trip}},\ D_{\mathrm{align}} - E_{\mathrm{trip}}].} \tag{16}$$

Every point of the interval is attained in the unrestricted free-phase comparison class.

Proof. Use $C_{\mathrm{eq}} = D - E_{\mathrm{trip}}$ and theorem 3.1. □

Remark 3.3 (Optimality and physical realizability)*.* The lower bounds in equations (15) and (16) apply to the literal packet because they use its actual moduli. Endpoint attainability is optimality in the larger class in which native phases may vary independently. Physical phases can be coupled by rows, masks, and divisor kernels, so an extremizing free-phase assignment need not be realized by the arithmetic packet. Therefore D_{dom} is the strongest universal modulus-only certificate, not a formula for the actual energy.

3.1 Comparison with the post-bin mass route

TPC-84's sharp modulus-only post-bin mass is

$$M_{\min}^{\mathrm{mag}} = \sum_n d_n.$$

Using all active determinant bins, TPC-83 then gives

$$D \geq \frac{(M_{\min}^{\mathrm{mag}})^2}{S}, \qquad S = \#\{n : A(n) \neq 0\}.$$

The direct energy certificate is never weaker.

Proposition 3.4 (Support-refined comparison)**.** *Let*

$$S_{\text{dom}} = \#\{n : d_n > 0\}.$$

Also write

$$S_{\text{actual}} = \#\{n : A(n) \neq 0\}, \qquad S_{\text{raw}} = \#\{n : F_n \neq \varnothing\}.$$

If $S_{\text{dom}} > 0$*, then*

$$D_{\text{dom}} \geq \frac{(\sum_n d_n)^2}{S_{\text{dom}}} \geq \frac{(\sum_n d_n)^2}{S_{\text{actual}}} \geq \frac{(\sum_n d_n)^2}{S_{\text{raw}}}, \tag{17}$$

where every displayed denominator is positive. Any of the three inequalities can be strict.

Proof. Cauchy–Schwarz on the positive defects gives the first inequality. For $d_n > 0$, the reverse triangle inequality gives

$$|A(n)| \geq d_n > 0.$$

Thus

$$S_{\text{dom}} \leq S_{\text{actual}} \leq S_{\text{raw}},$$

which gives the remaining comparisons. Strictness occurs, for example, when the positive defects are nonconstant or when either support inclusion is proper. □

The raw-fiber count is used in proposition 3.4; it is not substituted for the actual post-bin support in any TPC-83 norm identity.

3.2 A quantitative dominant-share theorem

Definition 3.5 (Uniformly dominant fibers)**.** For $0 < \varepsilon \leq 1/2$, let

$$\mathcal{G}_\varepsilon = \{n : R_n > 0, \quad r_{\max,n} \geq (1/2 + \varepsilon) R_n\}.$$

Write

$$R_{\mathcal{G}_\varepsilon} = \sum_{n \in \mathcal{G}_\varepsilon} R_n.$$

Theorem 3.6 (Dominant-share energy lower bound)**.** *For every finite packet and* $0 < \varepsilon \leq 1/2$*,*

$$\boxed{D \geq 4\varepsilon^2 \sum_{n \in \mathcal{G}_\varepsilon} R_n^2 \geq \frac{4\varepsilon^2 R_{\mathcal{G}_\varepsilon}^2}{\#\mathcal{G}_\varepsilon},} \tag{18}$$

where the second expression is omitted if $\mathcal{G}_\varepsilon = \varnothing$*.*

Proof. For $n \in \mathcal{G}_\varepsilon$,

$$d_n = 2r_{\max,n} - R_n \geq 2\varepsilon R_n > 0.$$

Consequently $\mathcal{G}_\varepsilon \subseteq \operatorname{supp} A$. Insert this in (15). The second inequality is Cauchy–Schwarz on $\mathcal{G}_\varepsilon$. □

Corollary 3.7 (Exponent transfer for the literal packet)**.** *Suppose, as additional hypotheses on the literal growing packet, that for fixed* $\mu, \kappa, \xi \geq 0$*,*

$$R_X \geq X^{-\mu+o(1)} N_{0,X}, \tag{19}$$

$$\varepsilon_X \geq X^{-\kappa+o(1)}, \tag{20}$$

$$R_{\mathcal{G}_{\varepsilon_X}} \geq X^{-\xi+o(1)} R_X. \tag{21}$$

Then

$$\boxed{D_X \geq X^{-2(\mu+\kappa+\xi)+o(1)} \frac{N_{0,X}^2}{Q_X}.} \tag{22}$$

Proof. The imported determinant interval gives $\mathcal{G}_{\varepsilon_X} \subseteq \operatorname{supp} A_X$ and hence $\#\mathcal{G}_{\varepsilon_X} \ll Q_X$. Insert (19)–(21) into (18). $\square$

Remark 3.8 (What is still missing). Corollary 3.7 proves an exponent-exact implication, not its hypotheses. In particular, the TPC-32 raw absolute-mass upper bound cannot replace (19). A finite census can measure D_{dom} and the dominant share at sampled scales, but a uniform lower bound for the growing literal packet remains an L2 problem.

Example 3.9 (Sharp phase-blind stop model)**.** Put two equal moduli in every fiber. Then $d_n = 0$ and $D_{\mathrm{dom}} = 0$, however large the raw mass is. Opposite phases make the actual energy zero, while aligned phases make it maximal. Thus a zero dominance certificate kills only the modulus-only route; it does not upper-bound the actual signed energy.

4 Exact shared-target rigidity

We return to the literal canonical parent set $\mathcal{T}_X$ of section 2. The positivity of the orbit coordinate and the signed canonical determinant together make target collisions much more rigid than a generic equality of integers.

Theorem 4.1 (Ordered composite-key uniqueness)**.** *Let*

$$t = (\alpha, \gamma, j), \qquad t' = (\alpha', \gamma', j')$$

be canonical parent keys with

$$\nu(t) = \nu(t'), \qquad x_t = x_{t'}, \qquad y_t = y_{t'}. \tag{23}$$

Then $t = t'$. Equivalently, the map

$$t \longmapsto (\nu(t), x_t, y_t) \tag{24}$$

is injective on the canonical parent archive.

Proof. The two target equalities imply

$$m_\alpha j = m_{\alpha'} j', \qquad m_\gamma j = m_{\gamma'} j'.$$

The ordered target pairs have the same gcd, say c. Equality of the normalized determinants therefore gives

$$m_\alpha - m_\gamma = m_{\alpha'} - m_{\gamma'} =: d.$$

The off-diagonal mask makes $d \neq 0$. Subtracting the two target equalities gives

$$dj = dj',$$

so $j = j'$. It follows that

$$m_\alpha = m_{\alpha'}, \qquad m_\gamma = m_{\gamma'}.$$

The physical row-slope uniqueness then gives $\alpha = \alpha'$ and $\gamma = \gamma'$. $\square$

Remark 4.2 (Canonical-parent premise). The theorem is a statement about mathematical parent keys. Two computational expansion rows may have the same parent; they must be inverse-aggregated before (24) is tested. Conversely, equality of the two complex numbers $u_t = u_{t'}$ without equality of their ordered targets is merely a coefficient coincidence and is not a physical duplicate in the present sense.

For the most inclusive target-sharing statement, allow equality between either slot of one ordered target pair and either slot of the other.

Definition 4.3 (Any-slot shared-target graph)**.** The graph $\mathscr{G}_{\mathrm{tar}}$ has vertex set $\mathcal{T}_X$. Distinct vertices t, t' are joined when

$$\nu(t) = \nu(t') \quad \text{and} \quad \{x_t, y_t\} \cap \{x_{t'}, y_{t'}\} \neq \varnothing.$$

Lemma 4.4 (A distinct pair shares at most one target). *Two distinct vertices of $\mathscr{G}_{\mathrm{tar}}$ satisfy exactly one of the four possible target equalities*

$$x_t = x_{t'}, \quad x_t = y_{t'}, \quad y_t = x_{t'}, \quad y_t = y_{t'}.$$

In particular, no off-diagonal edge is counted by two target-slot relations.

Proof. Because $j > 0$ and $m_\alpha \neq m_\gamma$, one has

$$x_t - y_t = (m_\alpha - m_\gamma)j \neq 0,$$

and similarly $x_{t'} \neq y_{t'}$. Thus two equalities that use the same target slot of one vertex would force the two targets of the other vertex to coincide.

It remains to consider two perfect matchings between the slots. The ordered matching

$$x_t = x_{t'}, \qquad y_t = y_{t'}$$

forces $t = t'$ by theorem 4.1. For the crossed matching

$$x_t = y_{t'}, \qquad y_t = x_{t'},$$

the contents agree. Since the determinants agree, the signed row differences agree:

$$m_\alpha - m_\gamma = m_{\alpha'} - m_{\gamma'} =: d \neq 0.$$

But target differences now give

$$dj = x_t - y_t = -(x_{t'} - y_{t'}) = -dj'.$$

Hence $j' = -j$, contradicting $j, j' > 0$. ☐

4.1 Divisor parametrization of one target group

For $s \in \{L, R\}$, write

$$T_L(t) = x_t, \qquad T_R(t) = y_t,$$

and define

$$G_s(k,T) = \{t \in \mathcal{T}_X : \nu(t) = k,\ T_s(t) = T\}, \qquad g_s(k,T) = \#G_s(k,T). \tag{25}$$

Proposition 4.5 (Exact candidate parametrization). *Fix $k \neq 0$ and a positive target T.*

(i) Every $t = (\alpha, \gamma, j) \in G_L(k,T)$ is determined by a pair of positive divisors

$$m_\alpha \mid T - h_0, \qquad c_t \mid T,$$

through

$$j = \frac{T - h_0}{m_\alpha}, \qquad m_\gamma = m_\alpha - kc_t.$$

(ii) Every $t \in G_R(k,T)$ is determined by

$$m_\gamma \mid T - h_0, \qquad c_t \mid T,$$

through

$$j = \frac{T - h_0}{m_\gamma}, \qquad m_\alpha = m_\gamma + kc_t.$$

The formulas are necessary candidate conditions. The physical row, primitive, support, mask, and exact-gcd tests may only remove candidates.

Proof. For the left group, $T = m_\alpha j + h_0$, so $m_\alpha j = T - h_0$. Also $c_t = (x_t, y_t) \mid x_t = T$, and $\nu(t) = k$ gives

$$m_\alpha - m_\gamma = kc_t.$$

The row slope determines the row key uniquely. These equations prove (i). The right-hand argument is symmetric. □

Corollary 4.6 (Uniform divisor bound)**.** *Uniformly in k, T, and the physical packet,*

$$\boxed{g_L(k,T),\ g_R(k,T) \le \tau(|T-h_0|)\tau(T) \ll_{\varepsilon,h_0,\mathscr{D}} X^{\varepsilon}.} \tag{26}$$

Proof. The first estimate counts the divisor pairs in proposition 4.5. On the physical support, $T \asymp_{\mathscr{D}} X$ and $T - h_0 \asymp_{\mathscr{D}} X$. The standard divisor bound $\tau(n) \ll_\varepsilon n^\varepsilon$ gives the last estimate [1]. □

Remark 4.7 (What was not used). No cancellation in u_t, no distribution of primes, and no average over h_0 enters proposition 4.5 and corollary 4.6. The estimate is a structural count on one prescribed shift. It is also only an upper bound for a collision class; it does not lower-bound the raw or post-bin mass.

5 Exact compressed shared-target census

The rigidity of lemma 4.4 makes the full shared-target correlation computable from group sums without a pair table. For $s \in \{L, R\}$, set

$$U_s(k,T) = \sum_{t\in G_s(k,T)} u_t, \qquad E_s(k,T) = \sum_{t\in G_s(k,T)} |u_t|^2. \tag{27}$$

All empty-group quantities are zero.

Let C_{tar} denote the ordered part of C_{eq} whose distinct vertices share at least one complete target in any slot. Split it by the unique slot relation from lemma 4.4.

Theorem 5.1 (Exact compressed target-correlation identity)**.** *For every finite canonical parent packet,*

$$\boxed{C_{\mathrm{tar}} = C_{LL} + C_{RR} + C_{LR},} \tag{28}$$

where

$$C_{LL} = \sum_{k,T} \Big(|U_L(k,T)|^2 - E_L(k,T)\Big), \tag{29}$$

$$C_{RR} = \sum_{k,T} \Big(|U_R(k,T)|^2 - E_R(k,T)\Big), \tag{30}$$

$$C_{LR} = 2\,\mathrm{Re} \sum_{k,T} U_L(k,T)\overline{U_R(k,T)}. \tag{31}$$

Each displayed sum is exact; no pair is deleted or counted twice.

Proof. For a fixed left group,

$$|U_L|^2 - E_L = \sum_{\substack{t,t'\in G_L \\ t\neq t'}} u_t \overline{u_{t'}},$$

which is the ordered same-left contribution. Sum over (k,T). The right contribution is identical.

A canonical off-diagonal parent never belongs to both $G_L(k,T)$ and $G_R(k,T)$, because $x_t \neq y_t$. Thus

$$U_L(k,T)\overline{U_R(k,T)}$$

contains precisely one orientation of every cross-slot pair at target T; adding its conjugate gives the ordered two-orientation sum. Finally, lemma 4.4 says that the four slot relations are disjoint on distinct vertices. Hence (29)–(31) have neither omissions nor overlaps. □

5.1 Exact degrees from group counts

Let $\deg_{\mathrm{tar}}(t)$ be the degree of t in $\mathscr{G}_{\mathrm{tar}}$, and let

$$\Delta_{\mathrm{tar}} = \max_{t\in\mathcal{T}_X} \deg_{\mathrm{tar}}(t).$$

Proposition 5.2 (Exact degree identity)**.** *For a vertex t with*

$$k = \nu(t), \qquad x = x_t, \qquad y = y_t,$$

one has

$$\boxed{\deg_{\mathrm{tar}}(t) = g_L(k,x) + g_R(k,x) + g_L(k,y) + g_R(k,y) - 2.} \tag{32}$$

Consequently,

$$\Delta_{\mathrm{tar}} \ll_{\varepsilon,h_0,\mathscr{D}} X^{\varepsilon}. \tag{33}$$

Proof. The neighbors sharing x consist of

$$G_L(k,x)\setminus\{t\} \quad \text{and} \quad G_R(k,x).$$

The neighbors sharing y consist of

$$G_R(k,y)\setminus\{t\} \quad \text{and} \quad G_L(k,y).$$

By lemma 4.4, these four neighbor lists are disjoint. Their cardinalities give (32). Equation (33) follows from corollary 4.6. ☐

Corollary 5.3 (Composite-key audit invariant)**.** *In a correctly inverse-aggregated archive:*

(i) the composite key (ν, x, y) occurs once;

(ii) every pair of distinct parent identifiers occurs in at most one target-slot join; and

(iii) the neighbor count produced by the joins equals (32)*.*

Failure of any item is an archive or canonicalization failure before it is evidence for arithmetic coherence.

Proof. The three assertions are respectively theorem 4.1, lemma 4.4, and proposition 5.2. ☐

5.2 Exact duplicates

Proposition 5.4 (No off-diagonal full-target duplicates)**.** *Suppose the phrase "exact physical duplicate" includes the canonical normalized determinant and the two ordered complete targets, as required by the native archive schema. Then after canonical parent aggregation its off-diagonal class is empty.*

Proof. Two such records have equal composite key (ν, x, y), so theorem 4.1 makes their parent keys equal. They are not an off-diagonal pair. ☐

Remark 5.5 (Coefficient coincidence is different)*.* Two distinct keys may happen to have the same numerical complex coefficient $u_t = u_{t'}$ while their targets differ. That equality does not satisfy proposition 5.4; it belongs to a separately defined coefficient-level relation. Calling it an exact physical duplicate without the target fields would weaken the schema and invalidate the uniqueness audit.

6 The loss-one shared-target barrier

The degree identity becomes an arithmetic scale bound through a finite graph inequality.

Lemma 6.1 (Degree–energy inequality)**.** *Let $\mathscr{G} = (V, E)$ be a finite simple graph of maximum degree Δ, and attach coefficients $u_v \in \mathbb{C}$. Then*

$$\boxed{\left| \sum_{\{v,w\} \in E} (u_v \overline{u_w} + u_w \overline{u_v}) \right| \leq \Delta \sum_{v \in V} |u_v|^2.} \tag{34}$$

The factor Δ is best possible.

Proof. For every edge,

$$|u_v \overline{u_w} + u_w \overline{u_v}| \leq 2|u_v||u_w| \leq |u_v|^2 + |u_w|^2.$$

Summing assigns $|u_v|^2$ exactly $\deg(v)$ times and proves (34). For sharpness, take the complete graph $K_{\Delta+1}$ and $u_v = 1$ for every vertex. The left side is $\Delta(\Delta + 1)$, while the energy is $\Delta + 1$. □

Remark 6.2 (Relation to spectral majorants)*.* Equation (34) is equivalently the adjacency operator bound $\|A_{\mathscr{G}}\|_{2\to 2} \leq \Delta$. The degree constant is sharp, but a large degree is only a necessary combinatorial capacity: it supplies no favorable sign for the actual coefficient. This is the same distinction between a nonnegative graph majorant and a signed operator emphasized in TPC-68 [5].

Theorem 6.3 (Shared-target coherence is at most loss one)**.** *For the literal fixed-h_0 matched packet,*

$$\boxed{|C_{\mathrm{tar},X}| \leq X^{o(1)} E_{\mathrm{trip},X} \leq X^{-1+o(1)} Q_X^3 J_X^2.} \tag{35}$$

Proof. Apply lemma 6.1 to $\mathscr{G}_{\mathrm{tar}}$, use the subpower degree (33), and then the imported triplet-diagonal bound (11). □

Corollary 6.4 (Conditional removal from every below-one main term)**.** *Assume that a future theorem proves, along a growing sequence,*

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2 \qquad \textit{for a fixed } 0 \leq \lambda_D < 1. \tag{36}$$

Then

$$|C_{\mathrm{tar},X}| = o(D_X), \qquad E_{\mathrm{trip},X} = o(D_X). \tag{37}$$

If

$$C_{\mathrm{other},X} = C_{\mathrm{eq},X} - C_{\mathrm{tar},X},$$

then

$$\boxed{C_{\mathrm{other},X} = (1 - o(1)) D_X > 0.} \tag{38}$$

Proof. Divide (35) and (11) by (36). The ratio is $X^{-1+\lambda_D+o(1)} = o(1)$. Now use

$$D_X = E_{\mathrm{trip},X} + C_{\mathrm{tar},X} + C_{\mathrm{other},X}.$$

□

Remark 6.5 (The hypothesis is essential)*.* This paper does not prove (36). Without it, (35) is an absolute upper bound and one cannot write $|C_{\mathrm{tar}}| = o(D)$, since D itself may vanish. The conditional quantifier in corollary 6.4 is therefore part of the theorem, not a stylistic caveat.

6.1 A general pairing-degree kill rule

Corollary 6.6 (Pairing-degree exponent barrier)**.** *Let $\mathscr{P}_X$ be any conjugation-closed simple pairing class on the canonical native keys, and let $C_{\mathscr{P},X}$ be its ordered coherence. If its maximum degree satisfies*

$$\Delta_{\mathscr{P},X} \le X^{\theta+o(1)}$$

for fixed $\theta \ge 0$, then

$$|C_{\mathscr{P},X}| \le X^{\theta-1+o(1)} Q_X^3 J_X^2. \tag{39}$$

Thus the class cannot by itself carry a lower bound with fixed loss $\lambda_D < 1-\theta$.

Proof. Use lemma 6.1 and (11). Comparison with the proposed lower scale gives the final assertion. □

Remark 6.7 (What survives the kill). The shared-target class has $\theta = 0$, so it is removed from every below-one main term. This does not treat:

(i) sharing a selected divisor while both complete targets differ;

(ii) a determinant-preserving symmetry with no target equality;

(iii) a broad signed remainder not described by a sparse graph; or

(iv) phase coupling that proves a direct lower bound for D_{dom} or D.

These are the honest remaining coherence routes.

6.2 Endpoint ledgers

If a future determinant lower bound has loss λ_D and a distinguished zero-mode estimate has gain η_Z, the inherited determinant compatibility remains

$$\lambda_D \le 2\eta_Z.$$

Separately, the full physical route must satisfy

$$\Lambda_{\mathrm{phys}} < \frac{1}{400}.$$

Equality in the second inequality is a stop. The loss-one shared-target estimate is a route-removal theorem; it is not a positive loss to be charged again in either ledger.

7 Audit-ready census and finite regression

The main theorems require no numerical experiment. A finite census is nevertheless useful for measuring the sharp dominance certificate and for deciding which remaining pairing relation deserves analysis. The schema below separates theorem inputs from diagnostics.

7.1 Required parent fields

For every scale X and fixed nonzero h_0, each canonical parent record stores:

1. experiment revision, parameter manifest, global $(Q_X, J_X, C_X, N_{0,X})$, and fixed-support identifiers;
2. canonical parent identifier (α, γ, j), exact row identifiers and slopes m_α, m_γ, and the positive orbit coordinate j;
3. exact targets x_t, y_t, content $c_t = (x_t, y_t)$, signed determinant numerator $m_\alpha - m_\gamma$, and exact quotient $\nu(t)$;

4. the final exact or directed-interval value of u_t, with the matched-channel and divisor expansion tree pointing back to this parent;
5. physical and computational multiplicities in different fields, together with the inverse-aggregation check; and
6. every active support, mask, orientation, cutoff, and provenance flag needed to reconstruct the literal coefficient.

The canonical serialization includes the exact integer composite keys

$$(\nu, x, y), \qquad (\nu, L, x), \qquad (\nu, R, y).$$

Hashes may detect file changes but do not replace the exact equality tests.

7.2 Derived statistics

One determinant sort computes, for every n,

$$R_n, \quad r_{\max,n}, \quad d_n, \quad A_X(n), \quad |A_X(n)|^2.$$

It returns

$$R_X,\ M_X,\ D_X,\ E_{\mathrm{trip},X},\ C_{\mathrm{eq},X},\ D_{\mathrm{dom},X}.$$

Two target-group sorts compute

$$g_s(k,T), \quad U_s(k,T), \quad E_s(k,T), \qquad s \in \{L, R\}.$$

Equations (28) and (32) then return

$$C_{\mathrm{tar},X}, \quad \Delta_{\mathrm{tar},X}, \quad C_{\mathrm{other},X} = C_{\mathrm{eq},X} - C_{\mathrm{tar},X}$$

without enumerating pairs. Comparison sorting costs

$$O(N_{\mathrm{rec}} \log N_{\mathrm{rec}})$$

and $O(N_{\mathrm{rec}})$ external storage, with bounded-memory merge runs when necessary.

7.3 Mandatory invariants

Every completed run verifies:

$$c_t = (x_t, y_t), \qquad c_t\nu(t) = m_\alpha - m_\gamma, \tag{40}$$
$$D_X - E_{\mathrm{trip},X} = C_{\mathrm{eq},X}, \qquad D_X \geq D_{\mathrm{dom},X}, \tag{41}$$
$$C_{\mathrm{tar},X} = C_{LL,X} + C_{RR,X} + C_{LR,X}, \qquad |C_{\mathrm{tar},X}| \leq \Delta_{\mathrm{tar},X} E_{\mathrm{trip},X}. \tag{42}$$

It also requires:

(i) uniqueness of (ν, x, y);

(ii) at most one target-slot equality for every distinct equal-determinant pair in a bounded direct regression;

(iii) agreement of the exact degree formula with direct neighbor counts in that bounded regression; and

(iv) agreement of the compressed correlation with a direct ordered pair sum.

The companion script `experiments/tpc95_certificate.py` checks these identities exactly on finite primitive integer packets. It is a regression test, not evidence for an asymptotic lower bound.

7.4 Decision rules

Dominance go. A theorem proving

$$D_{\mathrm{dom},X} \geq X^{-\lambda+o(1)} \frac{N_{0,X}^2}{Q_X}$$

immediately proves the same lower bound for D_X. The dominant-share hypotheses of corollary 3.7 are a sufficient route.

Dominance stop. If an exact analysis proves

$$D_{\mathrm{dom},X} \leq X^{-c+o(1)} \frac{N_{0,X}^2}{Q_X}$$

on an unbounded sequence for some fixed $c > 0$, then the modulus-only natural-dominance certificate fails on that sequence. The conclusion does not upper-bound D_X and does not kill signed phase structure.

Shared-target stop. Theorem 6.3 already stops the shared-target class as a main term for every fixed loss below one. No computation is needed for that exponent conclusion. A nonzero census value remains useful only as a finite normalization check or as a lower-order term.

Remaining-pairing go. For a new determinant-preserving relation, archive an exact conjugation-closed edge rule, prove or compute its degree without duplicating edges, and apply corollary 6.6. Passing the degree capacity test is necessary but not sufficient: a positive main term and a one-sided remainder estimate are still required.

Promotion boundary. Finite scaling plots for D_{dom}, C_{tar}, or a remaining class are L1 diagnostics. Promotion to L2 requires a uniform growing-X estimate for the actual fixed-h_0 packet, including all masks, short fibers, exceptional sectors, and the original global normalization.

8 Audited conclusion and next gate

Theorem 8.1 (Audited determinant-fiber conclusion)**.** *For every finite native determinant packet, let*

$$d_n = (2r_{\max,n} - R_n)_+.$$

Then

$$D \geq \sum_n d_n^2,$$

and this is the sharp lower endpoint over all phase assignments with the same fiber moduli. The corresponding free-phase coherence range is

$$\left[\sum_n d_n^2 - E_{\mathrm{trip}},\ \sum_n R_n^2 - E_{\mathrm{trip}}\right].$$

For the canonical parent TPC-32 packet at one prescribed nonzero h_0, the composite key

$$(\nu(t), x_t, y_t)$$

is injective. Distinct equal-determinant parents share at most one complete target in any slot. Their complete shared-target correlation has the exact compressed formula (28), and

$$|C_{\mathrm{tar},X}| \leq X^{-1+o(1)} Q_X^3 J_X^2.$$

Conditional on any future lower bound

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2 \quad \textit{with fixed } \lambda_D < 1,$$

one has

$$|C_{\mathrm{tar},X}| = o(D_X),$$

and the remaining non-target-sharing coherence carries $(1 - o(1))D_X$*.*

Proof. Combine theorems 3.1, 4.1, 5.1 and 6.3, corollaries 3.2 and 6.4, and lemma 4.4. □

Exact progress. The paper adds one positive and one negative certificate. The positive certificate is the exact modulus-only determinant-energy floor D_{dom}, with a sharp dominant-share exponent transfer. The negative certificate is a loss-one theorem for every same-determinant complete-target collision, including left–right cross-sharing. It converts the TPC-84 shared-target candidate from an open possible main term into a rigorously lower-order class on every future below-one energy branch.

Canonical duplicate rule. If an exact physical duplicate includes both ordered complete targets, then after canonical parent aggregation its off-diagonal class is empty. Repeated computational children must not be counted as native coherence. Equal numerical coefficient values on different targets are a different relation and receive no conclusion from the duplicate statement.

What is not proved. No lower bound is proved for

$$R_X, \qquad M_X, \qquad D_{\mathrm{dom},X}, \qquad D_X.$$

No upper bound is proved for the distinguished zero mode. No positive main term is obtained from a shared-divisor class with distinct targets, a determinant-preserving symmetry, or the unpaired remainder. Hence the paper contains no new L2 growing fixed-shift arithmetic estimate.

Level ledger. Theorem 3.1 and corollary 3.2 are L0 finite geometry. Theorems 4.1, 5.1 and 6.3 are L1 results on the literal fixed-shift carrier. The conditional statement $|C_{\mathrm{tar}}| = o(D)$ is asserted only under the future lower bound with $\lambda_D < 1$.

Independent endpoint ledgers. Any future determinant lower and zero-mode upper bounds still require

$$\lambda_D \leq 2\eta_Z.$$

All other physical fixed-power losses separately require

$$\Lambda_{\mathrm{phys}} < 1/400.$$

The same loss is never counted in both, and equality in the physical ledger is a stop.

Arithmetic claim boundary. Nothing is specialized from a general prescribed $h_0 \neq 0$ to $h_0 = 2$. The paper proves no unweighted Chowla estimate, no breach of the sieve parity barrier, no fixed-shift Hardy–Littlewood asymptotic, no prime-pair lower bound, and no twin-prime conclusion.

Next gate. Run the exact D_{dom} and compressed C_{tar} diagnostics on the literal archive. Analytically, the next positive theorem must either:

(i) prove a raw-mass lower bound together with a macroscopic dominant-fiber share;

(ii) find a high-capacity determinant-preserving pairing that does not share complete targets and control its signed remainder; or

(iii) prove the determinant-energy lower bound directly by a different arithmetic mechanism.

Shared-target coherence is no longer one of those doors.

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